

GANGCA, China

WMO: 527540

Lat: 37.3305N

Lon: 100.1380E

Elev: 10833

StdP: 9.78

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-3.0	-27.7	1.8	6.5	-23.9	2.2	6.8	27.3	23.9	24.3	23.2	4.7	0	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	68.4	52.9	65.6	51.4	63.0	49.9	55.4	65.1	53.7	62.5	52.1	60.3	7.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.9	86.6	60.4	50.1	81.2	58.8	48.5	76.1	57.3	28.6	65.5	27.3	62.8	26.0	60.7	61.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
23.9	20.0	16.4	DB	-12.7	73.2	3.6	3.0	-15.3	75.4	-17.3	77.2	-19.3	78.9	-21.9	81.1	
			WB	-14.0	57.6	3.3	1.8	-16.4	58.9	-18.3	60.0	-20.2	61.0	-22.6	62.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	11.3	16.9	25.6	36.0	43.0	48.7	53.6	53.0	46.2	35.6	24.1	15.1
	DBStd	15.44	5.53	6.39	6.47	6.10	4.79	4.06	4.40	4.51	4.61	5.06	5.60	4.69
	HDD50	6066	1200	927	756	420	223	69	12	19	135	448	776	1081
	HDD65	11251	1665	1347	1221	869	683	488	354	374	565	913	1226	1546
	CDD50	292	0	0	0	1	5	31	124	111	20	0	0	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	7	0	0	0	0	0	0	6	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	7.1	5.9	6.6	8.1	8.5	8.3	7.4	6.9	6.6	6.5	6.5	6.9	6.6
	PrecAvg	14.9	0.0	0.1	0.2	0.5	1.6	2.7	3.4	3.3	2.2	0.6	0.1	0.0
Precipitation	PrecMax	20.3	0.2	0.2	0.8	1.4	3.5	4.5	6.2	5.7	5.0	1.7	0.5	0.2
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.1	1.1	1.2	1.8	0.7	0.0	0.0	0.0
	PrecStd	2.5	0.0	0.0	0.2	0.4	0.8	0.9	1.0	1.0	0.9	0.4	0.1	0.0
	PrecStd	2.5	0.0	0.0	0.2	0.4	0.8	0.9	1.0	1.0	0.9	0.4	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	33.1	40.7	51.7	60.5	63.7	66.7	74.1	71.5	65.4	56.1	45.0	36.1	
	0.4% MCWB	21.8	26.3	32.5	37.8	42.0	49.4	55.1	54.5	52.3	39.9	30.3	24.6	
	2% DB	29.3	36.8	45.8	55.6	59.8	63.5	69.5	68.2	61.1	52.2	41.8	32.1	
	2% MCWB	19.6	24.4	29.8	36.5	41.0	48.1	53.6	53.1	49.3	37.7	28.6	21.6	
	5% DB	26.2	33.2	42.1	52.0	57.0	60.8	66.2	65.5	58.2	49.4	39.2	29.2	
	5% MCWB	17.6	22.3	27.8	34.8	40.0	46.6	51.9	52.3	46.7	35.7	27.0	20.0	
	10% DB	23.1	29.7	38.6	48.6	54.1	58.4	63.2	62.6	55.5	46.5	36.3	26.3	
	10% MCWB	15.7	20.3	26.2	33.3	39.0	45.4	50.7	50.9	44.8	34.4	25.4	18.1	
	0.4% WB	22.3	27.0	33.7	40.8	45.8	54.0	58.0	57.8	54.5	43.9	32.0	25.0	
	0.4% MCDB	32.1	39.3	48.2	54.6	57.3	62.9	69.1	67.5	61.9	52.7	42.2	34.9	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	19.8	24.6	30.6	38.2	43.7	50.8	55.6	55.6	51.5	40.1	29.4	22.2	
	2% MCDB	28.8	36.2	43.4	51.7	55.1	59.3	66.1	64.7	58.7	48.4	40.1	31.4	
	5% WB	17.8	22.6	28.7	36.3	42.0	48.8	53.8	54.0	49.0	37.7	27.6	20.2	
	5% MCDB	25.8	32.5	40.3	49.3	52.9	57.2	63.1	62.6	56.2	46.3	37.9	28.7	
	10% WB	15.9	20.5	26.8	34.5	40.4	47.1	52.1	52.2	46.6	35.6	25.8	18.3	
	10% MCDB	22.7	29.2	37.6	46.5	51.0	55.9	60.8	60.2	53.6	44.4	35.4	26.1	
Mean Daily Temperature Range	MDBR	23.1	23.7	23.0	20.6	18.5	16.7	16.4	16.7	17.0	20.4	22.7	22.9	
	5% DB	25.0	26.3	26.7	25.0	23.3	20.8	20.9	19.8	19.5	22.8	24.0	24.5	
	MCWBR	18.4	17.8	15.7	12.2	10.9	10.0	9.6	9.4	10.0	11.8	14.7	17.3	
	5% WB	24.3	25.3	23.6	21.8	18.9	16.5	16.7	16.1	16.4	19.2	22.6	23.5	
	MCWBR	18.0	17.3	14.1	11.3	10.0	9.1	8.5	8.4	9.0	11.0	14.3	16.8	
	MCWBR	18.0	17.3	14.1	11.3	10.0	9.1	8.5	8.4	9.0	11.0	14.3	16.8	
Clear-Sky Solar Irradiance	taub	0.212	0.249	0.304	0.357	0.373	0.374	0.363	0.358	0.325	0.275	0.228	0.203	
	taud	2.469	2.383	2.245	2.121	2.163	2.248	2.357	2.356	2.417	2.493	2.500	2.500	
	Ebn at Noon	322	317	305	292	288	286	289	288	294	303	312	319	
	Edh at Noon	25	31	40	49	48	44	39	38	34	28	24	23	
All-Sky Solar Radiation	RadAvg	987	1238	1588	1878	1964	1899	1878	1731	1464	1319	1065	921	
	RadStd	57	57	82	80	102	104	144	116	95	76	62	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.79	N/A	-2.14	N/A	N/A	N/A	-203	-288	+32	N/A
(47)	Regional (2 neighbors)	+0.59	N/A	-1.44	N/A	N/A	N/A	N/A	N/A	+94	N/A

Nomenclature: See separate page